arXiv:2607.15776v1 [cs.AI] 17 Jul 2026

NeurOWL: An LLM-Based Neural-symbolic Framework for Incomplete OWL Ontology Reasoning

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Abstract. OWL ontologies provide a formal knowledge representation framework that enables semantic reasoning, and have been widely adopted across domains such as healthcare and bioinformatics. In practice, however, real-world ontologies are often incomplete, which pose challenges for reasoning. In this work, we focus on a fundamental subsumption reasoning problem: given an incomplete ontology and a candidate (non-entailed) subsumption, determine whether the subsumption is semantically plausible and, if so, providing a logically sound explanation containing potential missing axioms. This task unifies subsumption verification with ontology abduction, and generalizes the latter by removing the need for a predefined candidate set of missing axioms. To address this subsumption reasoning problem, we propose NeurOWL, an end-to-end neuro-symbolic framework that jointly performs verification and abduction, leveraging both formally defined semantics and textual semantics through Large Language Models and ontology embeddings. We evaluate NeurOWL on real-world ontologies across multiple domains, demonstrating strong and robust performance across different domains.

1 Introduction

Ontologies expressed in the Web Ontology Language (OWL) provide a formal framework for representing structured knowledge and enable automated reasoning through Description Logics (DLs) [3]. Owing to their precise and shareable representation of machine-interpretable semantics, OWL ontologies are widely used in knowledge-intensive domains such as biomedicine [2] and healthcare [7].

A fundamental reasoning task in ontology engineering is subsumption reasoning: determining whether a subsumption relationship holds between two concepts and providing explanations for such entailments. For complete and consistent ontologies, subsumption can be determined by standard reasoners [17, 24], and explanations can be derived as *justifications* (i.e., minimal sets of axioms sufficient for the entailment) [1,25,32,42]. In practice, however, real-world ontologies are often incomplete due to different reasons like evolving domain knowledge

and the high cost of expert curation. As a result, some valid subsumptions may not be entailed because some required axioms are missing.

Abductive reasoning has been proposed as a natural solution to this problem [6, 11, 13–16, 18–20, 26–28, 33, 37]. In this work, we concentrate on *TBox abduction* that aims to find a set of hypotheses (missing axioms) $\mathcal{H}$ such that, when added to an incomplete ontology $\mathcal{O}'$, the target subsumption α becomes entailed (i.e., $\mathcal{O}' \cup \mathcal{H} \models \alpha$). Despite their promise, existing approaches exhibit several limitations: (1) they typically rely on a predefined hypothesis space, restricting candidate axioms to a fixed vocabulary [19, 28]; (2) they often assume that the target subsumption is valid, preventing their applicability in more general settings where the validity of α is itself uncertain. Recently, there has been growing interest in using large language models (LLMs) for related ontology reasoning and construction tasks such as subsumption prediction and concept placement [5, 8, 21, 22, 31, 34, 36, 39, 43]; however, these works generally lack explanatory capabilities.

To address these challenges, we propose `NeurOWL`, an **end-to-end** neuro-symbolic framework for explainable subsumption reasoning over incomplete OWL ontologies. By combing textual information and formal semantics through LLMs and ontology embeddings, `NeurOWL` removes the need for a predefined hypothesis space and supports both positive and negative entailment scenarios. Moreover, it operates in both training-free and fine-tuned settings, making it adaptable across domains.

The core idea of `NeurOWL` is to uncover missing knowledge by identifying intermediate concepts that connect related notions, as illustrated in Figure 1. Consider a simple example (see Example 2 for the full version) with the ontology $\mathcal{O}'_1$:

$$\{\texttt{PersianCat} \sqsubseteq \texttt{PetCat},\ \texttt{Cat} \sqsubseteq \texttt{Mammal},\ \texttt{Mammal} \sqsubseteq \texttt{Animal} \sqcap \exists \texttt{produces.Milk}\}.$$

Intuitively, one expects the subsumption $\texttt{PersianCat} \sqsubseteq \texttt{Animal}$ to hold, but it cannot be derived from $\mathcal{O}'_1$ due to incompleteness. To address this, `NeurOWL` searches for intermediate concepts C such that $\texttt{PersianCat} \sqsubseteq C \sqsubseteq \texttt{Animal}$, which can be used to construct missing axioms and explanations. Depending on the scenario, part or all of the connections may be missing (cf. Stages 2a/2b and 3a in Figure 1). For example, `NeurOWL` may select $C = \texttt{Cat}$ and construct the missing axiom $\texttt{PetCat} \sqsubseteq \texttt{Cat}$, while the other part, $\texttt{Cat} \sqsubseteq \texttt{Animal}$, is entailed by $\mathcal{O}'_1$ and is therefore not missing). Then, the explanation consist of the missing axioms and the axioms that entail $\texttt{Cat} \sqsubseteq \texttt{Animal}$. Candidate intermediate concepts are identified using ontology embeddings [39] and LLMs, which capture semantic similarity and verify likely missing connections. If no suitable intermediate concept is found, `NeurOWL` treats the target subsumption as a simple, indivisible axiom, proposes it directly, and validates it using an LLM (Stage 3b in Figure 1).

We evaluate `NeurOWL` on multiple real-world ontologies across diverse domains. Experimental results demonstrate that our approach achieves strong perfor-

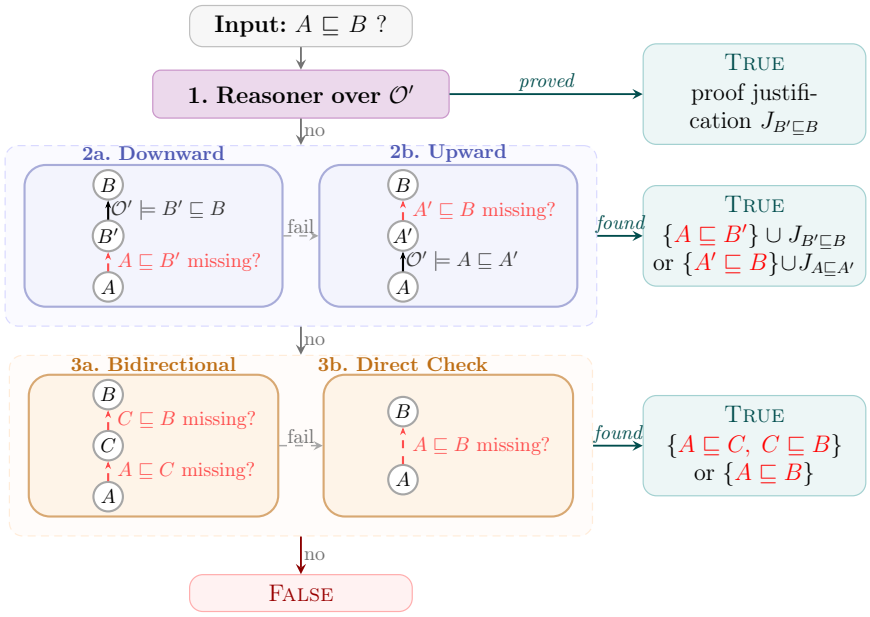


Fig. 1: Overview of NeurOWL: Red axioms denote potentially missing axioms.

mance in concept subsumption reasoning while also generating meaningful explanations for incomplete input ontologies. Our results shows that, on datasets constructed from the real-world ontologies FoodOn [12] and Snomed CT [7], NeurOWL achieves high performance in checking subsumption plausibility, with F1 scores of up to 97. They also demonstrate strong accuracy in explanations generated, as reflected in the correctness of predicted missing axioms, achieving X-F1 and X-F1* scores of up to 89.3 and 79.3, respectively. Further details are provided in Section 6.

The main contributions of this work are:

- We introduce a generalized TBox abduction task that does not require a predefined candidate set of missing axioms and handles cases where the given subsumption is incorrect.
- We propose NeurOWL, an end-to-end neuro-symbolic framework for reasoning over incomplete ontologies, utilising ontology embedding and LLM.
- We empirically evaluate our approach on real-world ontologies, demonstrating its effectiveness and robustness under varying degrees of incompleteness.

2 Related Work

Abductive Reasoning on ontologies can be divided into several categories, including *TBox abduction* [14,19,37], *ABox abduction* [11,13,15,18,20,26,27,33],

knowledge-base abduction [16,28], and *concept abduction* [6]. These categories are distinguished by the types of the missing elements: rules (TBox axioms), facts (ABox axioms), a combination of both, and concepts, respectively. In this work, we focus on TBox abduction, where the missing elements are TBox axioms, as introduced in Section 3.

Existing approaches to TBox abduction rely on different assumptions regarding the subsumption of a predefined set of candidate axioms. For example, [37] assumes a human-verified selection defined by an oracle function provided by domain experts; [14] relies on a set of plausible patterns; and [19,28] consider only axioms formulated over a given set of concepts and roles, known as signatures. These assumptions limit the practicality of such methods in real-world applications. Moreover, these approaches focus on logical computation and largely ignore the use of informal meta information like textual labels which also contain important semantics.

Ontology Embeddings aim to encode ontology entities (i.e., concepts, roles, and instances) as numerical vectors while preserving their structural and semantic properties. These embeddings support several downstream machine learning and data mining tasks including concept subsumption prediction [10]. Existing approaches can be broadly divided into two categories. (1) *Geometric model-based methods* [23, 29, 38, 40] represent ontology entities as geometric objects, translating description logic (DL) operators into geometric operations. For example, concept subsumption and intersection are modeled as region inclusion and intersection, respectively. (2) *Language-model-based methods* [9, 30, 35] leverage both formal knowledge and textual information to capture both linguistic and semantic features in the embedding space. More recently introduced, HiT [22] combines language models with hierarchical embeddings in hyperbolic space to represent taxonomies and support direct subsumption inference between two arbitrary concept labels. Building on this, OnT [39] extends the approach to $\mathcal{EL}$ ontologies by introducing role embeddings as rotations and additional loss functions to encode logical properties. In this work, we adopt OnT as our embedding method.

LLMs for Ontology Subsumption have been investigated in many tasks such as subsumption prediction or inference [8, 21, 22, 39, 43], taxonomy completion [34], axiom learning [5, 31, 43], and the intrinsic knowledge of hierarchical structures in LLMs [36]. However, to the best of our knowledge, few studies have specifically explored the use of LLMs for assessing the semantic plausibility of subsumptions. [44] is a related approach but assuming additional verification by human experts. Moreover, most of these LLM-based works, as well as the above introduced ontology embedding-based works, focus on the correctness of the subsumption, and little attention has been devoted to explanations. [41] investigates explanation generation and considers incomplete settings, but assume the correctness of the given subsumption and focus on evaluating the capabilities of LLMs instead of developing an end-to-end reasoner.

3 Preliminaries

Ontologies provide a formal framework for representing structured knowledge through collections of logical statements, known as *axioms*. These axioms describe relationships between *concepts* (unary predicates) and *roles* (binary predicates). Here, we take $\mathcal{EL}$ ontologies as an example, which is a lightweight yet expressive ontology that has been successfully employed in areas such as life sciences and semantic data integration [4]. Let $\mathsf{N_C} = \{A, B, \ldots\}$, $\mathsf{N_R} = \{r, s, \ldots\}$ denote two mutually disjoint sets of *concept names*, *role names*, respectively. The set of $\mathcal{EL}$*-concepts* is inductively defined as follows: $C ::= \top \mid A \mid C \sqcap D \mid \exists r.C$, where $A \in \mathsf{N_C}$, $r \in \mathsf{N_R}$, and C, D are $\mathcal{EL}$-concepts. In this work, we concentrated on TBox axioms, which also called *General Concept Inclusions* (GCIs) axioms of the form $C \sqsubseteq D$, where both C and D are $\mathcal{EL}$-concepts.

Example 1. Given the concept names `PersianCat`, `PetCat`, `Cat`, `Mammal`, `Animal`, `Milk` and the role `produces`, we may define a simple ontology $\mathcal{O}_1'$ as follows: $\{\texttt{PersianCat} \sqsubseteq \texttt{PetCat},\ \texttt{Cat} \sqsubseteq \texttt{Mammal},\ \texttt{Mammal} \sqsubseteq \texttt{Animal} \sqcap \exists \texttt{produces}.\texttt{Milk}\}$.

Logical reasoning over an ontology $\mathcal{O}$ enables the derivation of implicit knowledge from the stated axioms. The entailment relation "$\models$" is formally defined using interpretations and models (e.g., from $A \sqsubseteq B, B \sqsubseteq C \in \mathcal{O}$, one can infer $A \sqsubseteq C$, written as $\mathcal{O} \models A \sqsubseteq C$. See [4] for more details). To understand why an entailment holds, one can examine its *justifications*, which are minimal sets of axioms from which it follows [32].

Definition 1 (Justification). *Let $\mathcal{O}$ be an ontology such that $\mathcal{O} \models A \sqsubseteq B$. A* justification *for $A \sqsubseteq B$ is a subset $J_{A \sqsubseteq B} \subseteq \mathcal{O}$ that is minimal with respect to set inclusion and satisfies $J_{A \sqsubseteq B} \models A \sqsubseteq B$.*

4 The `NeurOWL` System

The task we wish to solve is the plausibility of a given concept inclusion $\alpha = A \sqsubseteq B$ being implied by a possibly incomplete ontology $\mathcal{O}'$, and return a set E of axioms as a possible explanation, i.e., We aim to determine the plausibility of given concept subsumptions over an $\mathcal{O}', E \models \alpha$. To ensure justifications rely primarily on the stated axioms in $\mathcal{O}'$ we limit the approach to return at most two missing Concept Inclusion axioms, if possible.

Without loss of generality, we focus on subsumptions of the form $A \sqsubseteq B$ between atomic concepts A, B. This setting generalizes naturally to subsumptions between complex concepts, $C \sqsubseteq D$, since a reasoner can reduce complex expressions by introducing fresh concept names (e.g., $A \equiv C$ and $B \equiv D$). Moreover, embeddings can be applied on complex concepts through verbalizations, as demonstrated in [39].

4.1 Main Structure

The overall NeurOWL framework is illustrated in Figure 1. It integrates logical structure with neural verification to predict missing axioms and consists of three stages:

- *Stage 1. Logical Checking*: Determine whether the input query can be logically entailed by $\mathcal{O}'$ using a standard ontology reasoner.
- *Stage 2. Logical Bridging*: Identify potentially missing axioms using candidate concepts induced by the logical structure of $\mathcal{O}'$, combined with LLM-based validation.
- *Stage 3. Bridging by Embedding*: Generate candidates directly based on semantic similarity using embeddings when the previous stages fail.

In **Stage 1** (Logical Checking), we use a standard description logic reasoner to test whether $\mathcal{O}' \models A \sqsubseteq B$ holds. If so, the system returns TRUE with a justification derived from $\mathcal{O}'$. Otherwise, NeurOWL assumes some axioms are missing and proceeds to subsequent stages.

In **Stage 2** (Logical Bridging), we attempts to identify *missing bridge concepts* using the logical information in $\mathcal{O}'$. There are two cases:

- *Stage 2a: Downward.* Consider each child B' of B in $\mathcal{O}'$ (i.e., $\mathcal{O}' \models B' \sqsubseteq B$) as candidate for a bridge concepts. Then, $A \sqsubseteq B'$ is treated as a candidate missing axiom and will be verified by the LLM.
- *Stage 2b: Upward.* Consider each parent A' of A in $\mathcal{O}'$ (i.e., $\mathcal{O}' \models A \sqsubseteq A'$) as candidate, and use the LLM to validate the corresponding potential missing axiom $A' \sqsubseteq B$.

Each missing axiom candidate is then verified by an LLM using prompt learning. If a valid bridge is found, the system returns TRUE along with the constructed justification. For example, if $A \sqsubseteq B'$ is validated, the justification is:

$$\{A \sqsubseteq B'\} \cup J_{B' \sqsubseteq B},$$

Note that LLMs may produce redundant answers that can be pruned directly. For instance, if B'_1 and B'_2 are selected in Stage 2a with $\mathcal{O}' \models B'_1 \sqsubseteq B'_2$, it suffices to retain B'_1 with missing axiom $A \sqsubseteq B'_1$, since $A \sqsubseteq B'_2$ follows from $\mathcal{O}' \cup \{A \sqsubseteq B'_1\}$. Example 2 provides a more realistic instance.

In implementation, instead of directly calling LLM to check each potential missing axiom, NeurOWL first uses an ontology embedding model (e.g., OnT [39]) to retrieve the top-k candidates, which are ranked by the subsumption score defined by the embedding model. This not only reduces the cost and computation time, but also utilises more semantics embedded.

In **Stage 3 (Bridging by Embeddings)**, if no suitable candidates are found in the previous stage, we search for missing axioms based purely on semantic information.

– *Stage 3a (bidirectional search):* We first perform a bidirectional search (Stage 3a in Figure 1) for an intermediate concept C such that $A \sqsubseteq C \sqsubseteq B$, where C is, by default, drawn from all atomic concepts in $\mathcal{O}'$ (excluding A and B), but can be customized based on user interest (e.g., existential restriction concepts of the form $\exists r.B_1$).

  We then select the top-k candidates and verify them using the LLM. Each candidate is ranked by the scoring function

$$\bar{s}(C) = \frac{1}{2}\big(s(A \sqsubseteq C) + s(C \sqsubseteq B)\big).$$

  If a valid C is identified, the system returns TRUE with the justification $\{A \sqsubseteq C,\ C \sqsubseteq B\}$. Otherwise, the subsumption $A \sqsubseteq B$ is treated as an *elementary assertion* (i.e., requiring no further decomposition) and as a potential missing axiom itself.

– *Stage 3b (Direct check):* Then, `NeurOWL` uses LLM to check the correctness of the subsumption $A \sqsubseteq B$ directly in Stage 3b, as shown in Figure 1. If the LLM judges $A \sqsubseteq B$ to be correct, `NeurOWL` returns TRUE along with the subsumption itself as the missing axiom and explanation. Otherwise, `NeurOWL` indicating that the input subsumption does not plausibly hold and returns FALSE.

Example 2 (Continuation of Example 1). The ontology $\mathcal{O}'_1$ given in Example 1 is incomplete and miss the axiom: $\texttt{PetCat} \sqsubseteq \texttt{Cat}$. Thus can not entail the following subsumption: $\texttt{PetCat} \sqsubseteq \texttt{Animal}$. If we input this subsumption into `NeurOWL`, than we have:

– **Stage 1:** The reasoner can not entail the given subsumption due to the incompleteness of $\mathcal{O}'_1$.

– **Stage 2a (Downward):** The system explores subclasses of `Animal`; in this simple case, only `Mammal` and `Cat` are considered, so no filtering step is required. This yields two candidate missing axioms: $\texttt{PetCat} \sqsubseteq \texttt{Cat}$, $\texttt{PetCat} \sqsubseteq \texttt{Mammal}$. A LLM is then used to select the axioms that are likely to be valid. If the LLM does not select any candidates, the process proceeds to the next stage.

– **Return:** Suppose the LLM identifies both candidate axioms as valid. In this case, `NeurOWL` concludes that the queried subsumption holds, yielding two outputs. Note that $\texttt{PetCat} \sqsubseteq \texttt{Mammal}$ can be ignored, as it is derivable from $\mathcal{O}'_1$ together with $\texttt{PetCat} \sqsubseteq \texttt{Cat}$. Finally, we output the missing axiom. $\texttt{PetCat} \sqsubseteq \texttt{Cat}$, and the corresponding explanation:

$$\{\texttt{PetCat} \sqsubseteq \texttt{Cat}\} \cup \{\texttt{Cat} \sqsubseteq \texttt{Mammal},\ \texttt{Mammal} \sqsubseteq \texttt{Animal}\},$$

  where the latter forms a justification for $\mathcal{O}'_1 \models \texttt{PetCat} \sqsubseteq \texttt{Animal}$.

It is worth noting that NeurOWL can be applied iteratively, using predicted missing axioms as new inputs (e.g., $\texttt{PetCat} \sqsubseteq \texttt{Cat}$ or $\texttt{PetCat} \sqsubseteq \texttt{Mammal}$ in the above example), to try to obtain increasingly fine-grained, higher-quality axioms. We discuss case studies to illustrate this potential in Section 6.4.

5 Dataset Construction

We construct a dataset from real-world ontologies to evaluate our method under two settings. datasets to evaluate our method under two settings.

Standard Setting In this case, we use all atomic concepts as the candidates set of bridge concept C (i.e, $A \sqsubseteq C \sqsubseteq B$ for given subsumption $A \sqsubseteq B$) of NeurOWL. The dataset is constructed as follows.

First, to simulate ontology incompleteness in a controlled manner, we construct a pruned ontology $\mathcal{O}'$ from the original ontology $\mathcal{O}$ by uniformly sampling and removing $r = 5\%$ of the TBox axioms. Then, the evaluation dataset with positive and negative samples of subsumptions are constructed as follows: **(1) Positive samples.** A positive sample is a subsumption $A \sqsubseteq B$ such that $\mathcal{O} \models A \sqsubseteq B$ but $\mathcal{O}' \not\models A \sqsubseteq B$. We construct the positive sample set $\mathcal{P}$ by randomly sampling from all positive candidates. That is, $\mathcal{P} \subseteq \mathcal{P}_{all} := \{(A, B) \mid \mathcal{O} \models A \sqsubseteq B \wedge \mathcal{O}' \not\models A \sqsubseteq B\}$. These positive samples may involve one or several bridge concepts, corresponding to downward, upward, or bidirectional cases associated with Stages 2a, 2b, and 3a, respectively. In such cases, the induced missing axioms, together with associated justifications for Stages 2a and 2b, serve as the ground-truth explanations. Alternatively, they may involve no bridge and are therefore expected to be resolved in Stage 3b; in this case, the subsumptions themselves serve as the ground truth explanation. **(2) Negative samples.** Negative samples are subsumptions $A \sqsubseteq B$ such that $\mathcal{O} \not\models A \sqsubseteq B$. To construct a balanced dataset, we sample an equal number of negative instances (i.e, $|\mathcal{N}| = |\mathcal{P}|$) by replacing B with a B' satisfying $\mathcal{O} \not\models A \sqsubseteq B'$, using the following two strategies:

- **Random:** randomly choose one B' from all possible candidates.
- **Hard:** restrict the candidates B' to those that are closer to A, in the sense that there exists a path of length at most 5 between A and B' in the taxonomy (ignoring the edge direction)[3]. If none exists, then choose B' randomly.

Complex Setting. Under standard settings, NeurOWL always returns subsumptions between atomic concepts as the missing axioms. However, in some cases, the missing axiom may involve a complex concept. For example, an ontology $\{A \sqsubseteq A_1,\ B_1 \sqsubseteq B_2,\ \exists r.B_2 \sqsubseteq B\}$ may miss plausible axiom $A \sqsubseteq \exists r.B_1$ which allows $A \sqsubseteq B$ to be derived.

To evaluate the performance of NeurOWL in such complex cases, we construct incomplete ontologies $\mathcal{O}'_{\exists}$ in which a bridge concept are of the form $\exists r.B_1$ for

[3] The taxonomy of $\mathcal{O}$ is a directed acyclic graph over its concepts, with an edge $A \to B$ iff $\mathcal{O} \models A \sqsubseteq B$ and no $B' \neq A, B$ satisfies $\mathcal{O} \models A \sqsubseteq B' \sqsubseteq B$.

Table 1: Dataset statistics (1,000 positive and 1,000 negative samples per dataset). $\#|\mathcal{P}_{\text{all}}|$ denotes the total number of possible positive samples. The "Splits" column reports the number of positive samples that can be solved in Stages 2a, 2b, and 3a, respectively; overlaps between stages may occur.

Dataset	**Concepts**	$\#$|Remove Axioms|	$\#\|\mathcal{P}_{all}\|$	**Splits** (2a/2b/3a/3b)
FoodOn$_A$	30,995	2,244 (5%)	102,872	813/777/977/23
Snomed$_A$	364,352	18,300 (5%)	1,181,331	651/569/897/103
Snomed$_\exists$	364,352	2,000 ($\exists$)	1,774	-/-/1000/0

some true non-entailed subsumptions $A \sqsubseteq B$. That is, $\mathcal{O}'_\exists \not\models A \sqsubseteq B$, while in the original ontology $\mathcal{O}$ we have $\mathcal{O} \models A \sqsubseteq \exists r.B_1 \sqsubseteq B$ for some B_1, and r.

Random removal is not suitable for constructing such incomplete ontologies, as it rarely yields the desired non-entailed subsumptions. Instead, we adopt a targeted pruning strategy. Specifically, we rank axioms containing existential restrictions (i.e., axioms in which $\exists r.B_1$ appears) by the number of direct subsumptions lost upon their removal, and select the most impactful ones to construct the pruned ontology $\mathcal{O}'_\exists$. Positive and negative samples are then generated in the same manner as before.

Note that, in this setting, we focus on evaluating the prediction of complex bridge concepts of the form $\exists r.B_1$. Therefore, `NeurOWL` is implemented using only Stages 1 and 3.

6 Evaluation

6.1 Experimental Setting

Datasets We use two real-world ontologies for our evaluation: FoodOn (2022-08-12 version) [12] and Snomed CT (2023-12-24 version) [7]. FoodOn captures general knowledge in the domain of food and agriculture, whereas Snomed CT provides detailed and specialized terms in the healthcare domain.

The datasets are obtained by implementing the data construction procedure described in Section 5 on both ontologies under two settings: (1) In the standard setting where atomic concepts serve as bridge concept candidates, we obtain two datasets, **FoodOn**$_A$ and **Snomed**$_A$. (2) For the complex case where we consider relation restriction concepts of the form $\exists r.\, B$ as candidates. This setting is applied only to Snomed CT, producing the dataset **Snomed**$_\exists$. We do not apply this procedure to FoodOn as it lacks subsumptions in which existential restrictions serve as bridges. The statistics of the generated datasets is shown in Table 1.

Implementation We tested two versions of `NeurOWL` with OnT[4] or pre-trained SBERT[5] as the embedding model, both using the `all-MiniLM-L12-v2` variant. While SBERT is used directly, OnT is fine-tuned[6] on the provided incomplete ontology $\mathcal{O}'$ for 20 epochs, with a learning rate of 10^{-5}, a training batch size of 256, a clustering loss margin of 3.0, and a centripetal loss margin of 0.5. We adopt `Qwen3.5-9B`[7] as the LLM in our experiments, because it is open-source, which makes work easier to reproduce, and achieves strong performance despite its relatively small size. Unless stated otherwise, we use non-thinking mode, as it delivers both high performance and fast inference speed.

Without affecting correctness, for a given subsumption $A \sqsubseteq B$, we restrict the candidate set to direct children of B in Stage 2a [8] and direct parents of A in Stage 2b. This restriction reduces the search space and improves prediction accuracy. Results using all parents/children of A/B are reported in Appendix A.1.

Baselines To the best of our knowledge, no existing methods directly fit our setting. We therefore introduce three simple baselines that assess only the plausibility of given subsumptions and do not generate explanations. The baselines are: (1) **LLM-only**, which uses the same prompt (see Figure 7 in the Appendix) as in Stage 3b to evaluate the plausibility of subsumptions; and (2) two embedding-based methods, **SBERT** and **OnT**, which classify subsumptions as true or false by assessing whether their similarity (for SBERT) or subsumption (for OnT) score exceeds a threshold. The threshold is selected to maximize the F1-score (defined below) on a validation set randomly sampled from the test set, containing 50 positive and 50 negative samples. For fair comparison, we use the same LLM and the same SBERT and OnT as in `NeurOWL`.

As discussed in Sections 1 and 2, existing abductive reasoning approaches are not included in our comparisons, as they differ fundamentally from our setting: they focus on the logical procedure of extracting possible missing axioms from a predefined hypothesis space and treat the given subsumption as an actual observation.

Evaluation Metrics We use the following three metrics:

- **F1**: The standard F1-score that measures the correctness of the True/False prediction, independent of explanations.
- **X-F1**: Requires both correct prediction and correct explanation. Specifically, for each positive sample: (1) if a ground-truth bridge concept exists (Stage 2a/2b/3a), the prediction is correct only if the model outputs `True` and return one of the ground-truth concepts; (2) otherwise, the prediction

[4] `https://github.com/HuiYang1997/OnT`
[5] `https://sbert.net/`
[6] using package at `https://pypi.org/project/ontology-transformer/`
[7] `https://huggingface.co/Qwen/Qwen3.5-9B`
[8] A is a direct children of B in $\mathcal{O}'$ if $\mathcal{O} \models A \sqsubseteq B$ and there is no A' such that $\mathcal{O} \models A \sqsubseteq A' \sqsubseteq B$. Direct parent is defined in the same way.

Table 2: Overall results on all three datasets.

Dataset	Method	Fine-tuned	Random Neg			Hard Neg		
			F1	X-F1	X-F1*	F1	X-F1	X-F1*
FoodOn_A	OnT	✓	0.910	—	—	0.663	—	—
	SBERT		0.712	—	—	0.514	—	—
	LLM-only		0.836	—	—	0.789	—	—
	`NeurOWL` (OnT)	✓	**0.960**	**0.893**	**0.793**	**0.845**	**0.781**	**0.684**
	`NeurOWL` (SBERT)		**0.960**	0.885	0.788	0.844	0.772	0.680
Snomed_A	OnT	✓	0.943	—	—	0.692	—	—
	SBERT		0.927	—	—	0.565	—	—
	LLM-only		0.772	—	—	0.734	—	—
	`NeurOWL` (OnT)	✓	0.968	**0.847**	**0.659**	**0.862**	**0.745**	**0.569**
	`NeurOWL` (SBERT)		**0.970**	0.846	0.646	0.856	0.736	0.550
$\text{Snomed}_\exists$	OnT	✓	0.897	—	—	0.612	—	—
	SBERT		**0.927**	—	—	0.623	—	—
	LLM-only		0.881	—	—	0.854	—	—
	`NeurOWL` (OnT)	✓	0.924	**0.497**	**0.193**	**0.872**	**0.460**	**0.176**
	`NeurOWL` (SBERT)		0.906	0.002	0.000	0.862	0.002	0.000

is correct only if the model outputs `True` at Stage 3b. For each negative sample, the prediction is correct if the model outputs `False`.

– **X-F1***: the same as X-F1 but requires all the output bridge concepts are among the ground-truth concepts in (1) in testing the positive sample.

Baseline methods are evaluated only using F1, as they cannot predict missing axioms or provide explanations.

6.2 Main Results

Summary The overall results are presented in Table 2. We observe that both variants of `NeurOWL` consistently outperform all baselines in terms of the standard F1 score. Specifically, in the hard negative setting on Snomed_A, `NeurOWL` (OnT) achieves an F1 score of 0.862, compared to 0.734 for the strongest baseline. This improvement can be attributed to the structure of `NeurOWL`, which enables it to effectively leverage information from the existing ontology (i.e., Stages 2 and 3), and thus leads to better performance in determining the plausibility of given subsumptions. There is only one exception, which occurs in the random negative setting on $\text{Snomed}_\exists$, where SBERT slightly outperforms both variants of `NeurOWL` in terms of F1. This may be due to the candidates of the relational restriction concept $\exists r.B_1$ being extremely large (i.e., beyond 6,000,000 combinations with 20 roles and over 300,000 concepts), which leads to poor retrieval performance of the possible bridged concepts. This is also reflected in

Table 3: Part 1 candidate ranking detailed breakdown (random negatives, positive samples only, %).

Dataset	Variant	Stage	H@1	H@5	H@10	H@100	MRR
FoodOn$_A$	NeurOWL (OnT)	Stage-2a	0.546	0.964	0.996	1.000	0.744
		Stage-2b	0.920	1.000	1.000	1.000	0.960
		Stage-3a(only)	0.429	0.714	0.714	0.857	0.538
	NeurOWL (SBERT)	Stage-2a	0.483	0.892	0.940	1.000	0.676
		Stage-2b	0.932	1.000	1.000	1.000	0.965
		Stage-3a(only)	0.286	0.857	0.857	0.857	0.476
Snomed$_A$	NeurOWL (OnT)	Stage-2a	0.664	0.949	0.974	1.000	0.786
		Stage-2b	0.886	1.000	1.000	1.000	0.937
		Stage-3a(only)	0.182	0.409	0.591	0.727	0.292
	NeurOWL (SBERT)	Stage-2a	0.528	0.886	0.951	0.997	0.679
		Stage-2b	0.903	1.000	1.000	1.000	0.948
		Stage-3a(only)	0.318	0.591	0.682	0.955	0.462
Snomed$_\exists$	NeurOWL (OnT)	Stage-3a	0.001	0.052	0.122	0.577	0.042
	NeurOWL (SBERT)	Stage-3a	0.000	0.000	0.000	0.001	0.000

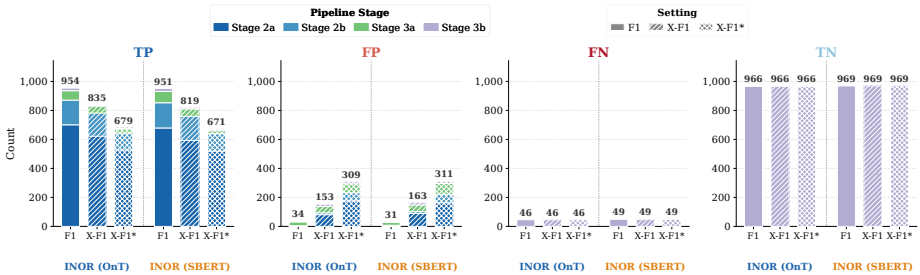
the relatively low X-F1 and X-F1* scores (<0.5) and the retrieval performance reported in Table 5 in the next section. However, NeurOWL still shows superior performance in the more challenging hard negative setting on the same dataset, Snomed$_\exists$, which further demonstrates the effectiveness of our method.

We also find that fine-tuning on the target ontology leads to consistent improvements: NeurOWL (OnT) always outperforms NeurOWL (SBERT). The performance gap is relatively small on simpler datasets such as FoodOn$_A$ and Snomed$_A$, where only atomic concepts are considered as candidates. In contrast, for more complex datasets like Snomed$_\exists$, which involve existential concepts of the form $\exists r.B$, NeurOWL (OnT) shows significantly better performance due to its adaptation to the underlying logical structure through fine-tuning. This suggests that, in relatively simple default settings, it is sufficient to implement NeurOWL in a training-free manner using the SBERT model. However, when considering more complex cases involving complex concepts, it is essential to use fine-tuned models such as OnT.

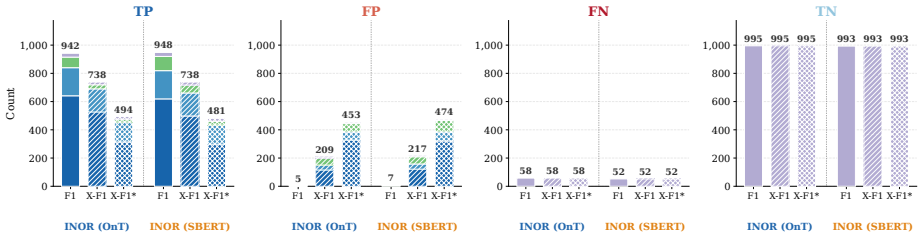
6.3 Detailed Analysis

Retrieval performance The retrieval results are summarised in Table 3. We categorise the positive samples into different groups based on whether they have downward/upward bridge concepts in Stage 2a/2b (which may overlap) or only bidirectional bridge concepts in Stage 3a (distinct from the former cases).

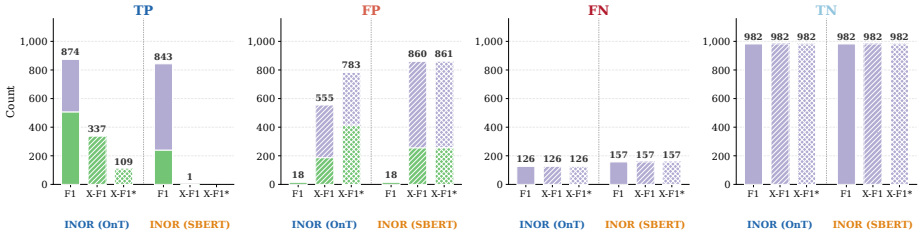
We observe that the retrieval performance of Stage 2a/2b is substantially higher than that of Stage 3a alone. This is reasonable, as the number of candidates in



(a) FoodOn$_A$



(b) Snomed$_A$



(c) Snomed$_\exists$

Fig. 2: Detailed results across different stages.

Stage 2a/2b (i.e., direct children/parents of B/A) is significantly smaller than the space of all atomic concepts. Furthermore, Stage 2b achieves better performance than Stage 2a due to the smaller number of direct parents compared to direct children (e.g., 1.2 vs. 32.7 on average in Snomed$_A$).

Moreover, we find that, due to the large candidate space of relational restriction concepts $\exists r.B_1$, the retrieval performance on the Snomed$_\exists$ dataset is considerably lower than on the other datasets. By fine-tuning on given incomplete ontologies, OnT achieves much better performance than SBERT; however, there remains a large gap compared to the default case where only atomic concepts are considered as the bridge.

Per-stage performance Figure 2 presents a detailed breakdown of the results at each stage, including true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN).

Table 4: Stage contribution ablation. "✓" indicates that the stage is applied, while "–" indicates that it is not. The default setting is shaded.

Dataset	2a	2b	3a	3b	Random-Neg			Hard-Neg		
					F1	X-F1	X-F1*	F1	X-F1	X-F1*
FoodOn_A	✓	–	–	–	0.823	0.766	0.689	0.789	0.733	0.656
	–	✓	–	–	0.832	0.811	0.701	0.771	0.751	0.645
	–	–	✓	–	0.779	0.650	0.367	0.723	0.600	0.334
	✓	–	–	✓	0.941	0.773	0.698	0.870	0.708	0.634
	–	✓	–	✓	0.917	0.818	0.710	0.837	0.740	0.637
	–	–	✓	✓	0.900	0.656	0.378	0.821	0.588	0.332
	✓	✓	✓	✓	**0.960**	**0.893**	**0.793**	**0.845**	**0.781**	**0.684**
Snomed_A	✓	–	–	–	0.781	0.689	0.478	0.743	0.653	0.448
	–	✓	–	–	0.718	0.667	0.611	0.673	0.622	0.569
	–	–	✓	–	0.672	0.556	0.356	0.634	0.524	0.335
	✓	–	–	✓	0.922	0.742	0.548	0.856	0.681	0.497
	–	✓	–	✓	0.884	0.721	**0.669**	0.819	0.660	**0.611**
	–	–	✓	✓	0.848	0.607	0.422	0.787	0.562	0.388
	✓	✓	✓	✓	**0.968**	**0.847**	0.659	**0.862**	**0.745**	0.569
$\text{Snomed}_\exists$	–	–	✓	–	0.463	0.156	0.060	0.447	**0.149**	**0.059**
	–	–	✓	✓	**0.911**	0.156	0.060	**0.873**	0.145	0.057

For both FoodOn_A and Snomed_A, the majority of TP predictions are generated in Stage 2a, followed by Stages 2b and 3a. In contrast, most FPs originate from Stage 3a. This observation aligns with the retrieval performance reported in Table 3, where Stage 3a exhibits relatively lower performance compared to Stages 2a and 2b. Furthermore, we observe that `NeurOWL` (SBERT) produces almost no TP predictions under the X-F1 and X-F1* metrics, while `NeurOWL` (OnT) achieves much better performance, indicating the superiority of fine-tuning in handling complex concepts.

Ablation of different stages Table 4 reports the performance obtained when using only subsets of the full pipeline. We find that relying on only one of Stages 2a, 2b, or 3a (i.e., rows marked by a single ✓) generally leads to weaker performance, particularly for Stage 3a, where the X-F1* score on FoodOn_A (random-neg) drops sharply from 0.793 to 0.367. For Snomed_A and $\text{Snomed}_\exists$, combining Stages 2b and 3b yields higher X-F1* scores than using the full pipeline. This may be due to the smaller candidate set obtained by focusing solely on direct parents, which improves the likelihood of achieving full correctness for all predicted bridge concepts. Despite this, the full pipeline still achieves better overall F1 and X-F1 scores, highlighting its stronger overall performance.

6.4 Case Studies

Single-iteration case We present two illustrative cases: one true positive (TP) and one false positive (FP). Assuming the given subsumption is always $A \sqsubseteq B$.

1. **(TP)** In Snomed_A, in the case $A =$ `Irritant contact blepharoconjunctivitis` and $B =$ `Disorder of soft tissue`. `NeurOWL` correctly identifies the subsumption $A \sqsubseteq B$ by discovering the bridge concept $C =$ `Dermatitis of eyelid` in Stage 2b. The LLM-only baseline incorrectly classifies this subsumption as false, likely due to the large semantic gap between A and B.
2. **(FP)** In FoodOn_A, in the case $A =$ `Grammatorcynus` and $B =$ `swine food product`. `NeurOWL` incorrectly identifies $C =$ `vertebrate animal food product` as a valid bridge concept in Stage 3a. This is erroneous, as the subsumption $C \sqsubseteq B$ does not hold.

Multi-iteration case Here, we consider iteratively applying `NeurOWL` as mentioned in Section 4, where the output missing subsumption in one iteration is used as the target subsumption in the next iteration. On Snomed_A, assume the initial input is $A \sqsubseteq B$ with $A =$ `Primary undifferentiated carcinoma of anterior wall of nasopharynx` and $B =$ `Disorder of nasopharynx`, and `NeurOWL` is performed by four iterations:

- **Iteration 1**: For input $A \sqsubseteq B$, `NeurOWL` outputs $C_1 =$ `Lesion of nasopharynx` in Stage 2a, together with the new missing axiom $A \sqsubseteq C_1$.
- **Iteration 2**: We input the missing axiom $A \sqsubseteq C_1$ obtained in Iteration 1 to `NeurOWL` again, and it produces $C_2 =$ `Neoplasm of anterior wall of nasopharynx` in Stage 2a, along with the new missing axiom $A \sqsubseteq C_2$.
- **Iteration 3**: With the input $A \sqsubseteq C_2$, `NeurOWL` produces $C_3 =$ `Malignant tumour of anterior wall of nasopharynx` in Stage 2a, together with the new missing axiom $A \sqsubseteq C_3$.
- **Iteration 4**: With the input $A \sqsubseteq C_3$, `NeurOWL` produces $C_4 =$ `Primary malignant neoplasm of anterior wall of nasopharynx` in Stage 2c, along with two new missing axioms $A \sqsubseteq C_4$ and $C_4 \sqsubseteq C_3$.

In summary, the four iterations of `NeurOWL` produces the following subsumption chain: $A \sqsubseteq C_4 \sqsubseteq C_3 \sqsubseteq C_2 \sqsubseteq C_1 \sqsubseteq B$, where all the identified missing axioms are correct and exhibit progressively finer granularity. This example demonstrates the potential of iterative applications of `NeurOWL` to yield increasingly precise and fine-grained results.

7 Conclusion

In this work, we proposed `NeurOWL`, an end-to-end framework for explainable neural-symbolic reasoning over incomplete OWL ontologies. By integrating the logical structure and textual information of the ontology via embeddings and

LLM, NeurOWL can infer the plausibility of a concept subsumption axiom and in addition can return a set of axioms as its justification, preferring stated axioms in the ontology when possible with up to two predicted missing axioms. Comprehensive experiments and case studies on three datasets based on real-world ontologies have demonstrated an overall good performance of NeurOWL and the effectiveness of its modules. In future work, we aim to extend NeurOWL to support more complex concepts with logical operators of $\forall$ and $\neg$ beyond $\exists$ considered in this work. We also would like to construct datasets and evaluate its iterative capabilities.

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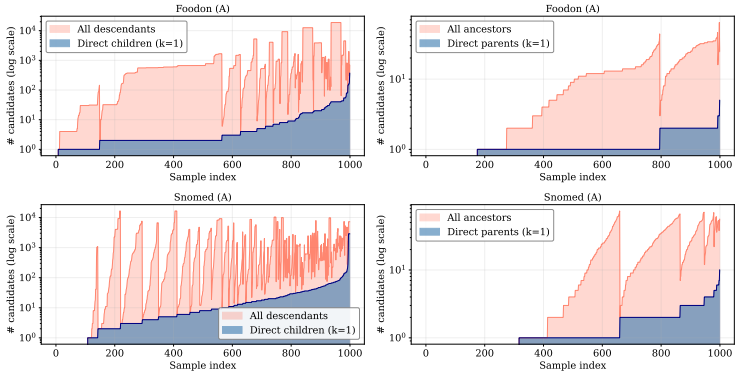


Fig. 3: Comparison between the number of direct and the all children and parents. Samples are ordered by increasing number of direct parents/children.

A Extra Results

A.1 Direct Children/Parents vs. All

In this section, we compare the performance impact of using only direct parents/children as candidates in Stage 2a/2b (denoted as *DIRECT*, the default setting of `NeurOWL`) versus using all transitive parent/child nodes (denoted as *ALL*).

We first present statistics for the number of direct and transitive parent/child nodes associated with the evaluated positive samples. As shown in Figure 4, the average number of direct parents is approximately 1–1.2, which is substantially lower than the average number of direct children (8.6–32.7). This observation aligns with the structural properties of ontologies: a concept typically has only a few direct parents but may have many direct children.

Furthermore, Figure 3 illustrates that the number of transitive (*ALL*) parents/children can be significantly larger than the number of direct ones, in some cases increasing by up to two orders of magnitude. This highlights the substantial expansion of the candidate space when transitive relationships are considered.

In Table 5, we present the retrieval performance under the *DIRECT* and *ALL* settings. As expected, the *DIRECT* setting consistently outperforms the *ALL* setting across all cases, demonstrating the advantage of restricting candidates to direct parents/children.

In Table 6, we report the overall performance of the *DIRECT* and *ALL* settings. While the *ALL* setting occasionally achieves higher F1 scores in certain cases, the *DIRECT* setting demonstrates superior overall performance. This advantage is particularly evident in the consistently higher X-F1 scores across all cases.

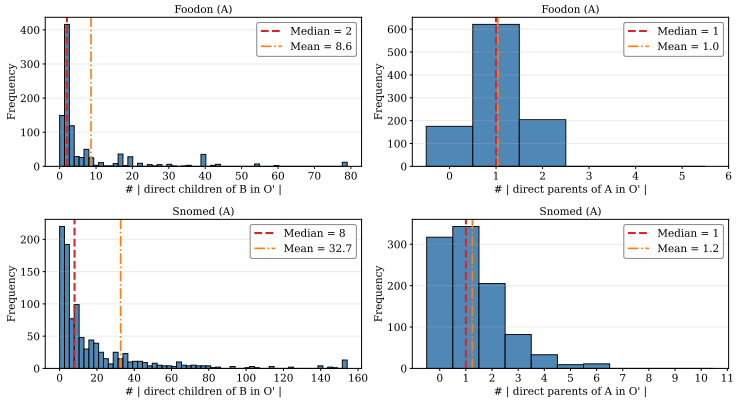


Fig. 4: Distribution of the number of direct children and parents for positive samples in test sets.

Table 5: Comparison of retrieval performance using using direct or all Parent/Child. All the results is shown in (*DIRECT*/*ALL*).

Dataset	`NeurOWL`	**Stage**	**H@1**	**H@5**	**H@10**	**H@100**	**MRR**
FoodOn_A	(OnT)	2a	**54.6** / 17.1	**96.4** / 39.2	**99.6** / 51.4	**100.0** / 82.9	**74.4** / 28.4
		2b	**92.0** / 87.9	**100.0** / 96.3	**100.0** / 98.2	100.0 / 100.0	**96.0** / 91.3
	(SBERT)	2a	**48.3** / 8.2	**89.2** / 17.8	**94.0** / 24.8	**100.0** / 61.0	**67.6** / 14.1
		2b	**93.2** / 82.5	**100.0** / 92.3	**100.0** / 97.4	100.0 / 100.0	**96.5** / 87.0
Snomed_A	(OnT)	2a	**66.4** / 30.6	**94.9** / 67.1	**97.4** / 79.7	**100.0** / 98.3	**78.6** / 47.2
		2b	**88.6** / 54.5	**100.0** / 77.5	**100.0** / 87.0	100.0 / 100.0	**93.7** / 64.8
	(SBERT)	2a	**52.8** / 24.4	**88.6** / 51.2	**95.1** / 61.6	**99.7** / 87.1	**67.9** / 36.9
		2b	**90.3** / 55.2	**100.0** / 83.1	**100.0** / 91.2	100.0 / 100.0	**94.8** / 67.0

However, for the X-F1* metric, the *ALL* setting outperforms *DIRECT* in some scenarios, such as the hard negative setting in *Ours (SBERT)*. This may be because the candidates in the *DIRECT* setting are highly relevant and difficult to distinguish, which can lead the LLM to mistakenly verify them as true bridge concepts. As a result, the X-F1* score decreases, since this metric requires all identified bridge concepts to be correct; such errors therefore reduce overall performance.

A.2 Ablation of LLMs, embedding models, and retrieval size

To better understand the contribution of different parts in `NeurOWL`, we conduct a systematic ablation study along three dimensions: (1) the base embedding model used for OnT; (2) the LLM employed in the verification stage; and (3) the candidate retrieval size k.

For the embedding model, we evaluate two alternatives:

Table 6: Overall results of *DIRECT* vs. *ALL*. All the results is shown in (*DIRECT*/*ALL*).

Dataset	NeurOWL	Random Neg			Hard Neg		
		F1	X-F1	X-F1*	F1	X-F1	X-F1*
FoodOn_A	(OnT)	96.0/**96.8**	**89.3**/87.6	**79.3**/75.0	84.5/**86.9**	**78.1**/78.0	**68.4**/65.8
	(SBERT)	**96.0**/95.3	**88.5**/83.9	**78.8**/75.8	84.4/**85.5**	**77.2**/74.3	**68.0**/66.8
Snomed_A	(OnT)	**96.8**/95.7	**84.7**/77.6	**65.9**/64.1	86.2/**87.1**	**74.5**/69.5	**56.9**/**56.9**
	(SBERT)	**97.0**/96.2	**84.6**/80.4	64.6/**66.9**	85.6/**87.0**	**73.6**/71.6	55.0/**59.0**

Table 7: Ablation study results on FoodOn_A. Default setting is shaded.

Variant	Embed	LLM (Qwen3.5)	k	Random Neg		Hard Neg	
				F1	X-F1	F1	X-F1
(1) Embed	MiniLM-L6	9B	10	**0.966**	**0.907**	0.856	0.799
	MPNet	9B	10	0.963	0.901	**0.858**	0.799
	MiniLM-L12	9B	10	0.964	0.905	0.857	**0.800**
(2) LLM	MiniLM-L12	9B (think)	10	0.907	0.862	0.855	**0.808**
	MiniLM-L12	4B	10	0.953	0.865	0.825	0.740
	MiniLM-L12	9B	10	**0.964**	**0.905**	**0.857**	0.800
(3) Top-k	MiniLM-L12	9B	5	**0.966**	0.900	**0.860**	0.794
	MiniLM-L12	9B	20	0.964	**0.907**	0.855	0.799
	MiniLM-L12	9B	10	0.964	0.905	0.857	**0.800**

– `all-MiniLM-L6-v2` (384-dimensional, same as `all-MiniLM-L12-v2` used by default) and

– `all-mpnet-base-v2` (768-dimensional).

For the LLM, we compare `Qwen3.5-9B` in thinking mode and `Qwen3.5-4B` (non-thinking) against the default `Qwen3.5-9B` (non-thinking). For retrieval, we vary the number of candidates using $k \in \{5, 20\}$, compared to the default setting of $k = 10$.

The results are summarized in Table 7. We can see that, overall, variations in the embedding model and the choice of k lead to only minor performance differences. The former suggests that all evaluated embedding models are sufficiently expressive for our datasets, while the latter is consistent with the strong retrieval performance reported in Table 3, indicating that changing k has limited impact on the final results.

In contrast, the choice of LLM has a more noticeable effect. Specifically, the thinking mode slightly reduces performance on random negative cases, but yields a slight improvement in X-F1 for hard negative cases.

LLM Prompt

```
You are an ontology expert. Your task is to determine subsumption
(is-a) relationships between concepts. Answer only YES or NO for each
question. Do not explain.

We want to determine: Is "{A}" a subclass of the following concepts:
- {candidate_1}
- {candidate_2}
...

For each candidate, answer YES if "{A}" is a subclass of it,
otherwise NO.

1. Is "{A}" a subclass of "{candidate_1}"?
2. Is "{A}" a subclass of "{candidate_2}"?
...

Answer format (one per line):
1. YES/NO
2. YES/NO
```

Fig. 5: Prompt template used for Stage 2a. The prompts for Stage 2b/3a follow a similar procedure, but with different candidate sets: candidates are parents of "A" for Stage 2b, or all atomic concepts for Stage 3a. The questions ask whether the candidates are subclasses of "B" for Stage 2a, or both directions for Stage 3a.

B Prompts

The prompts used for the LLMs in Stages 2a/2b and 3a setting are illustrated in Figures 5 and 6, respectively. Figure 5 focuses on the downward stage (i.e., Stage 2a), while Figure 6 shows the prompt used for handling complex cases, where role restrictions of the form $\exists r.\, B$ are treated as candidate sets in Stage 3a. The prompt for Stage 3b is illustrated in Figure 7.

LLM Prompt

```
You are an ontology expert analyzing concept relationships.

Query: Is "{A}" a subclass of "{B}"?

Below are {n} candidate existential restrictions that might justify
this classification. Each is of the form "role some Filler".


For each candidate, answer YES if:
1. "{A}" plausibly has this role-filler property
2. This filler captures the essential characteristic that defines
   membership in "{B}"

{numbered list of candidates}

Respond in the format:
1. YES/NO
2. YES/NO
...
Only output the numbered YES/NO answers, nothing else.
```

Fig. 6: Prompt template for Stage 3a with relation restriction concepts $\exists r.B$ as candidates. `{A}` and `{B}` are replaced with concept labels; `{n}` with the number of candidates; the numbered list enumerates each candidate as "`r some B`".

LLM Prompt

```
You are an ontology expert in the biomedical and food domains.
Determine whether the following subsumption relationship holds.

Is "{A}" a subclass of "{B}"?

Answer with only YES or NO.
```

Fig. 7: Prompt template for Stage 3b and LLM-only baseline.